METOPROLOL SUCCINATE

[Toprol XL]



Extended-release, cardioselective beta-1 blocker.
Lowers heart rate and blood pressure; protects the heart.

1 MECHANISM OF ACTION (MOA) & TARGET



- TARGET: Beta-1 adrenergic receptors
- Blocks catecholamine binding.
- ↓ heart rate + ↓ contractility.
- ↓ myocardial oxygen demand.



ANALOGY: A governor limiting engine RPMs to prevent overheating.

2 FORMS & STRENGTHS



Extended-Release (ER) Tablets:

- 25 mg
- 50 mg
- 100 mg
- 200 mg

3 INDICATIONS (FDA-APPROVED & MAJOR OFF-LABEL)



FDA-APPROVED:

- Hypertension
- Angina pectoris
- HFrEF (reduces mortality)

MAJOR OFF-LABEL:

- Atrial fibrillation rate control
- Migraine prophylaxis

4 DOSAGE & ADMINISTRATION

A) ADULT DOSE



- Standard: 100 mg daily
- Max: 400 mg daily
- HFrEF: Start 12.5–25 mg daily; double every 2 weeks to target 200 mg
- Angina: 100 mg daily

B) PEDIATRIC DOSE



- Age ≥6 years: 1 mg/kg daily
- Max: 2 mg/kg or 200 mg
- Route: PO

C) ADMINISTRATION



- Route: Oral (PO)
- Take consistently with or immediately following meals to enhance bioavailability

D) ADJUSTMENTS



- Renal: None required
- Hepatic impairment: initial dose; titrate cautiously

5 PHARMACOKINETICS & DYNAMICS (PK/PD)

ONSET	PEAK EFFECT	t½ (HALF-LIFE)	METABOLISM
			
~1 hour	2–4 hours	3–7 hours	Hepatic (CYP2D6)

6 ADVERSE EFFECTS & TOXICITY



COMMON:

- Fatigue, bradycardia, dizziness, depression, sexual dysfunction, cold extremities

SEVERE:

- Heart block,
- cardiogenic shock,
- bronchospasm,
- masking of hypoglycemia

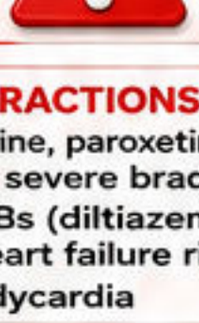
7 CONTRAINDICATIONS & INTERACTIONS

ABSOLUTE CONTRAINDICATIONS



- Severe bradycardia
- 2nd/3rd degree AV block
- Cardiogenic shock

BLACK BOX WARNING



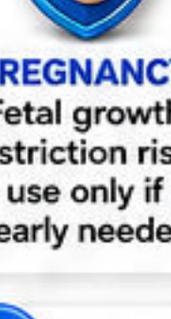
Exacerbation of ischemic heart disease upon abrupt withdrawal.

MAJOR DRUG INTERACTIONS

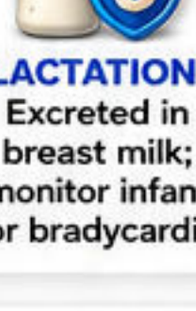


- CYP2D6 inhibitors (fluoxetine, paroxetine) → ↑ metoprolol levels → severe bradycardia
- Non-dihydropyridine CCBs (diltiazem, verapamil) → additive AV block & heart failure risk
- Digoxin → additive bradycardia

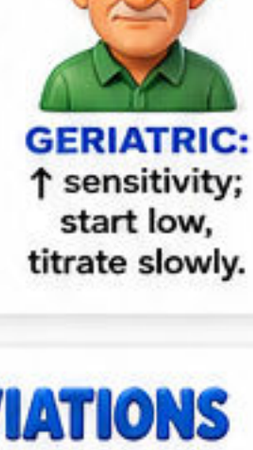
8 SPECIAL POPULATIONS (PREGNANCY & LACTATION)



PREGNANCY: Fetal growth restriction risk; use only if clearly needed.



LACTATION: Excreted in breast milk; monitor infant for bradycardia.



GERIATRIC: ↑ sensitivity; start low, titrate slowly.

9 CLINICAL PEARLS

- One of three beta-blockers proven to reduce HFrEF mortality (carvedilol, bisoprolol).
- Do not initiate during acute decompensated HF; patient must be euvolemic first.
- Taper over 1–2 weeks to prevent rebound ischemia or arrhythmias.
- Cardioselectivity is dose-dependent and lost at higher doses.
- Masks tachycardia in hypoglycemia.

10 GLOSSARY OF ABBREVIATIONS

AV	= Atrioventricular
CCB	= Calcium channel blocker
CYP2D6	= Cytochrome P450 2D6
ER	= Extended-release
HFrEF	= Heart failure with reduced ejection fraction